

## **Machine Learning Based Chatbot for Mfano Bora Africa Web Integration Information Requirement Analysis**

This is a detailed information gathering and system analyst research report for the Machine Learning-Based Web-Integrated Chatbot for Mfano Bora Africa. This analysis was carried out to ensure that all the information requirements of the chatbot are identified so that its deployment and integration on the Mfano Bora site becomes seamless. The report identifies the users of the chatbot as well as what they are expected to do so that they can be guided to either the right page, contact or answer by whoever accesses the chatbot. In addition, the report provides an open-source technology stack of pre-trained LLMs such as Groq API, Hugging Face Models and python libraries which will ensure evolution, adaptability and minimal information hallucinations. Lastly, the report identifies the entire system architecture through the identification of the PostgreSQL database schema, the API driven data pipeline with an admin dashboard for KPIs, the experience workflow of the floating chatbot module as well as the project file structure.

## 1. Information Requirements Analysis & Project Purpose

### Purpose

The main aim of this project is to create a smart, event-based chatbot to serve as a front-line digital assistant for Mfano Bora Africa Ltd. The chatbot will quickly respond to the questions raised by visitors, direct applicants to the attachment/career portal page, provide information about logistics consulting services, and integrate flawlessly into the current website interface.

### Information Requirements

To ensure that the chatbot gives accurate and factual information, the knowledge base should intake the below organizational data strictly:

- Company and Contact Information: Location (Mfano House, Ole Sein Rd, Nairobi), business hours (Monday-Friday: 7:00 AM – 5:00 PM, Saturday: 8:00 AM – 1:00 PM) and email (info@mfanoborafrica.com).
- Services Provided: Information about logistics and integrated supply chain management consultancy services and mission statements.
- Achievements & Activities: Information about East Africa Transport, Logistics & Road Safety Awards (24 years old) and Road Safety Club Kenya.
- Career Opportunities & Training Programs: Ways to join the industrial attachment (ICT, logistics, administrative), internship, mentoring and computer software packages, directly linking to the application portal.

## 2. User Roles and Personas

Identifying the user profiles makes sure that the conversation design solves particular problems and directs every question to its own solution or web page.

- Corporate Clients / Logistics Opportunities: Potential users looking to enhance operations, those requiring logistics consultation, or those looking to know more about industry standards within the company.
- Students & Alumni: Potential candidates applying for industrial attachments, internships, or computer applications.
- Award Nominees & Industry Colleagues: Site visitors looking for more information on E.A. Regional Awards, those looking at the gallery of award winners, or those interested in event timetables.
- Community Members: Potential members looking to become part of the Road Safety Club for national awareness activities and school campaigns.
- System Administrators: Mfano Bora employees charged with the responsibility of monitoring KPIs and the knowledge base from chat logs.

### 3. Technology Stack & AI Architecture

To ensure consistent learning while fully preventing hallucinations, a Retrieval-Augmented Generation (RAG) architecture is going to be employed. This approach would limit the LLM to generate answers from the Mfano Bora knowledge base only.

- Large Language Model (LLM): Groq API based on a very high-performance open-source model such as Llama-3 (e.g., Llama-3.3-70b) for extremely fast inference and reasoning capabilities.
- NLP & Embeddings: HuggingFace Sentence Transformers (Python) to create vector embeddings from website-scraped information and FAQs for semantic search.
- Backend & API Routing: Node.js with Express.js/Python (FastAPI) for handling stateless API requests and webhooks.
- Vector Database: PostgreSQL with `pgvector` extension or ChromaDB for storing and retrieving documents chunks based on user similarity scores.
- Frontend UI: React.js with integration of Tailwind CSS for creating a lightweight responsive floating chat widget.
- Security & Admin Auth: Firebase Authentication with RBAC.

#### 4. PostgreSQL Database Schema

All user activity will be logged in a PostgreSQL database to enable administrators to measure the quality of information and identify common inquiries, among other things.

##### Users schema

Captures unique visitor session data.

The key columns will include;

user\_id (UUID, PK), session\_token, created\_at, ip\_hash

##### chat\_sessions schema

Groups messages into active conversations.

The key columns will include;

session\_id (UUID, PK), user\_id (FK), start\_time, end\_time, status

##### chat\_logs schema

Stores exact conversation history.

The key columns will include;

log\_id (PK), session\_id (FK), sender\_type (User/Bot), message\_text, timestamp, intent\_matched

##### knowledge\_base schema

The SSOT (Single Source of Truth) for the bot.

The key columns will include;

doc\_id (PK), category, content\_chunk, embedding\_vector (pgvector), last\_updated

##### admin\_metrics schema

Pre-calculated KPIs for the dashboard.

The key columns will include;

metric\_id (PK), total\_conversations, fallback\_rate, resolution\_rate, date\_recorded

## 5. Data Pipeline & API Workflow

Data pipeline connects chatbot interface with machine learning and admin dashboard

1. Data Ingestion & Structuring: Verified data (FAQs, services, meta-data) is processed, split into chunks and loaded into the `knowledge\_base` database by Python data pipeline.
2. Query handling (API): User sends a request `POST /api/v1/chat/query`. Backend translates the query into vector representation and searches `pgvector` for 3 most relevant Mfano Bora context chunks.
3. LLM Generation: Context chunks and query are sent to Groq API. Instructions from guardrails indicate: "You are the Mfano Bora AI Assistant. Respond using only the context. If the answer cannot be found in the context, give the contact details."
4. Logging & Analytics: Backend calls asynchronously `POST /api/v1/logs/save` and logs the query to PostgreSQL.
5. Admin Dashboard (API): `GET /api/v1/admin/dashboard-stats` retrieves aggregated statistics on session durations, missed intents (missing information), and user metrics for the React.js admin interface.

## 6. User Experience (UX) Workflow

1. Initialization: The user is presented with a floating button called "MB" on the bottom-right corner of the website.
2. Greeting: Clicking the button causes the chat window to open, displaying the online status, followed by a prompt saying: "👋 Hello! I'm the AI Assistant for Mfano Bora Africa. I'm here to assist with services, attachments, location, and much more. What would you like to know?".
3. Interaction and Actionable Routing: In response to a student asking: "How do I apply for an attachment?", the bot searches for the information and provides the link: `[https://www.mfanoborafrica.com/attachment#career-form`](https://www.mfanoborafrica.com/attachment#career-form)
4. In case of a logistics query from a client, the bot summarizes the consultancy model and gives the link to the contact page.
5. Fallback Protocol: In case the question is vague or not found in the knowledge base, the bot responds with: "I cannot seem to find that exact piece of information. Feel free to contact our team via email [info@mfanoborafrica.com](mailto:info@mfanoborafrica.com) or visit Mfano House from Monday to Friday 7 AM-5 PM."

## 7. Project File Structure

For ease of deployment, maintenance, and scaling, the repository structure of the project will be modular as follows:

mfano-bora-chatbot-system/





